Simulated Drives over Measured Road Profiles for Car-Based IRI Estimation: What Transfers and What Needs Calibration

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***Abstract*—Estimating the International Roughness Index (IRI) from the vibration of ordinary cars needs paired data: drives of the car type over road sections with surveyed profiles. Vehicle simulation can generate such pairs from the profiles alone. We test what estimators trained on simulated drives transfer to a real car, using public profilometer-referenced data from one car model (LiRA-CD, four routes, 168 km), held-out routes and 5 km-block bootstrap intervals. Gradient boosting trained on real drives reaches a within-route mean absolute error (MAE) of 0.39 m/km. Trained only on a randomised half-car simulation over the measured profiles of the other routes, it ranks segments within routes about as well under each of nine sensor and vehicle models, but so does, within 0.05 in Spearman correlation on every route, an untrained, speed-normalised vibration RMS. What has to be learned is the level: the zero-shot bias ranges from -0.09 to $+0.43$ m/km and depends on the simulated sensor path, through the features carrying vibration above 10 Hz, and on the simulated vehicle population. With 5 km of profiled road and its paired drives, a level offset (pooled MAE 0.42–0.56 m/km) or joint training on simulated and real passes (0.45–0.47 m/km) beat training on those drives alone (0.68 m/km) for every variant, even with one pass per segment. With 20 km, joint training stayed better for every variant, whereas real drives alone beat the level offset for the three variants with the largest vibration deficit at 10–15 Hz. Simulated vibration also grows faster with speed than the real car's. For this car, simulation over profiled roads replaced most paired drives; whether this holds for other vehicles requires new paired data.**

***Index Terms*—International Roughness Index, road roughness, vehicle vibration, connected vehicles, simulation-to-real transfer, calibration.**

I. Introduction

The International Roughness Index (IRI) is the standard summary of longitudinal road roughness [1], [2]. Measuring it requires a longitudinal profile from a profilometer, so minor roads are surveyed rarely. Production cars that already carry accelerometers could fill the gap [3], [4], [5], but their vibration depends on the vehicle, the sensor mounting and the speed, so the mapping from vibration to IRI has to be established for each car type from paired data: drives of that car over road sections whose profile was measured [6]. Collecting such paired drives is the expensive step.

Vehicle simulation offers another source of pairs. Driving a vehicle model over profiles that road agencies already survey yields vibration with exact IRI labels, at any speed and for many vehicle parameter sets. Estimators trained this way have been validated in simulation [7], [8] and tested on real vehicles for the weighted longitudinal profile [9], for profile reconstruction [10] and for roughness classes [11]; calibrated vehicle models can invert real car vibration into road profiles [12], [13] or set an IRI–vibration relation by simulation [14]. What is missing is evidence, for continuous IRI from a production car on held-out roads of public data, of which parts of a simulation-trained estimator transfer, which simulator choices set its level, and how much paired real data must accompany the simulation.

We study this question on LiRA-CD [15], public data of one car model on four routes with profilometer references, comparing estimators trained on real cars with the same estimators trained only on simulated drives over the measured profiles of the other routes. We vary the sensor and vehicle model of the simulator, combine it with k km of profiled road with paired drives in three ways, and compare simulated and real drives at equal profile budget, in their speed dependence on the same real segments and in their vibration spectra.

The answer has three parts. Ranking segments within a route discriminates little: simulation-trained gradient boosting ranks them about as well as the real-trained model under every simulator variant, and an untrained, speed-normalised vibration RMS comes within 0.05 of both on every route. What has to be learned is the level, and it does not transfer reliably: it depends on the simulated sensor path, through the features carrying vibration above 10 Hz, and on the simulated vehicle population. A few kilometres of paired drives fix it: with 5 km, simulation over all profiled roads beats training on those drives alone for every variant, whether the drives set a level offset or enter joint training; with 20 km, joint training still does, whereas a level offset falls behind real drives alone for the variants with the largest vibration deficit at 10–15 Hz. Speed dependence transfers only in part, and the simulated vibration scales with speed differently from the real car's.

Our contributions are: (i) a route-disjoint benchmark on public profilometer-referenced car data with dependence-aware inference and baselines from response-type physics

(a calibrated quarter-car inversion), learned estimators and simulation-trained estimators (Section V-A); (ii) a decomposition of what simulation transfers to a real car across nine sensor and vehicle models, with a sensor factorial, a same-segment spectral comparison and a band-limited retraining that associate the level error with the simulated vibration and test which inputs carry it (Sections V-B and V-C); (iii) the paired real data that simulation replaces, at equal profile budget and with a level offset, an affine map or joint training, for every variant, with test-route intervals and the effect of the pass density (Section V-D); (iv) fixed-road speed elasticities of real- and simulation-trained estimators measured on real roads (Section V-E). The study revises an earlier, unpublished pipeline of ours trained on a game engine; its label and protocol problems are documented in Supplement S2, and its corpus serves here only as a negative reference. We will release the code, protocols, configurations and all predictions with the published version of the paper.

II. Related Work

Response-type measurement and physical vehicle models. IRI is defined on the longitudinal profile through a reference quarter car (the "Golden Car") driven at 80 km/h [1], [16], [2]. Response-type systems measure a vehicle's reaction instead of the profile and must be correlated with a reference profiler for each vehicle [6]. Model-based methods estimate roughness from the responses of a specific vehicle with fixed or identified vehicle models and transfer functions [17], [18]; Bridgelall's road impact factor normalises vertical-acceleration energy for connected vehicles [19]. Wu et al. [14] update a full-car model of the test vehicle from a few impact tests, derive a linear relation between IRI and vibration RMS by simulation over random profiles, and validate it on real roads; this calibrates the vehicle model instead of the estimator. Closest to our data, Skar and Levenberg [12] invert road profiles from the accelerations of ten nominally identical electric cars with a quarter-car model, a PID controller and nonlinear error minimisation, and compare them with a laser profilometer; Agebjär et al. [13] identify a vehicle model of a production car and reconstruct the profile with a Kalman filter. These methods need a model of each vehicle with known or identified parameters; our baseline M7 follows the same idea with a quarter car calibrated on the training routes, but it is not a reproduction of [12], whose calibrated vehicle parameters are not available (its open code contains only the Golden-Car parameters).

Learned estimators on real data. Smartphone studies report agreement with reference roughness that depends on speed, vehicle and phone [20], on mounting [21] and on the number of repeated runs [22]; see also [23], [3], [24]. Data-driven models accounting for speed, vehicle type and mounting were trained on real smartphone data [4], semi-supervised learning exploits unlabelled responses of crowdsourced vehicles [25], [26], and Jeong and Jo [27] validated a deep IRI estimator on data of 29 vehicles against an inertial profiler. On LiRA data, Bajic et al. [28] trained regressors on the car's vertical acceleration and speed along the M3 motorway and tested them on the last 20 % of the same road; we test on routes not used for training and report uncertainty that accounts for spatial dependence.

Learning from simulated responses. Closest in design to our study, Nitsche et al. [9] trained regressors on features of simulated vehicle responses and validated them with a probe vehicle, for the weighted longitudinal profile rather than IRI; according to Bajic et al. [28], the simulation used a measured road profile, the reference was an inertial profilometer and the effect of varying speed was not included. Networks trained on simulated responses have also been used to reconstruct profiles and classify roughness [29], then, trained on a full-vehicle model driven over slightly modified measured profiles, applied to the measured responses of that vehicle [10], to classify roughness grades from quarter-car simulations [8] and, with a real-vehicle test, from the unsprung-mass response [11], and to estimate IRI without vehicle-specific calibration: Jeong et al. [7] trained CNNs on simulated responses of four vehicle types with real driving-speed profiles and validated them numerically with real road IRI information. Randomising the simulator [30], [31] and adapting the randomisation with a few real roll-outs [32] are the standard devices for simulation-to-real transfer; calibration or joint training with paired drives and the same-segment spectral comparison play this role here. Game engines provide vehicle dynamics for software testing [33], [34], but IRI labels derived from their authored road meshes need validation (Supplement S2). We add, for continuous IRI from a production car on held-out roads of public data, a decomposition of what a simulation-trained estimator transfers, which simulator choices set its level, and how much paired data must accompany the simulation, for several ways of combining simulated and real data.

III. Data and Simulation

Real cars: LiRA-CD. LiRA-CD [15], [35] contains Green-Mobility Renault Zoe cars instrumented with an AutoPi telematics unit (three-axis accelerometer, GPS) connected to the CAN bus and fixed to the frame in the middle console, close to the front axle [15], and surveys of the same lanes by the Danish Road Directorate's P79 profilometer. We use the four routes with car data: two urban roads in Copenhagen (CPH1, part of the O2 ring road, and CPH6) and two motorways (M3, M13), each in both directions. The P79 files provide the wheel-path laser elevations at 0.1 m with GPS; the P79 surveyed CPH1, M3 and M13 in September 2020 and CPH6 in June 2020, and the car passes on these routes were recorded between November 2020 and August 2021, two to fifteen months later [35]. In the car files the accelerometer is logged at a nominal 52.6 Hz (66 % of passes) or 333 Hz, quantised to 0.01 g in most passes, with a median mounting tilt of 10°; speed comes from the CAN bus. After map matching (Section IV) the data contain 16,293 segment-passes over 1,678 distinct 100 m segments (168 km) from 91 passes of at least 8 cars (telematics units). Urban segments have mean IRI 3.36 m/km and are driven at 43 km/h on average, motorway segments

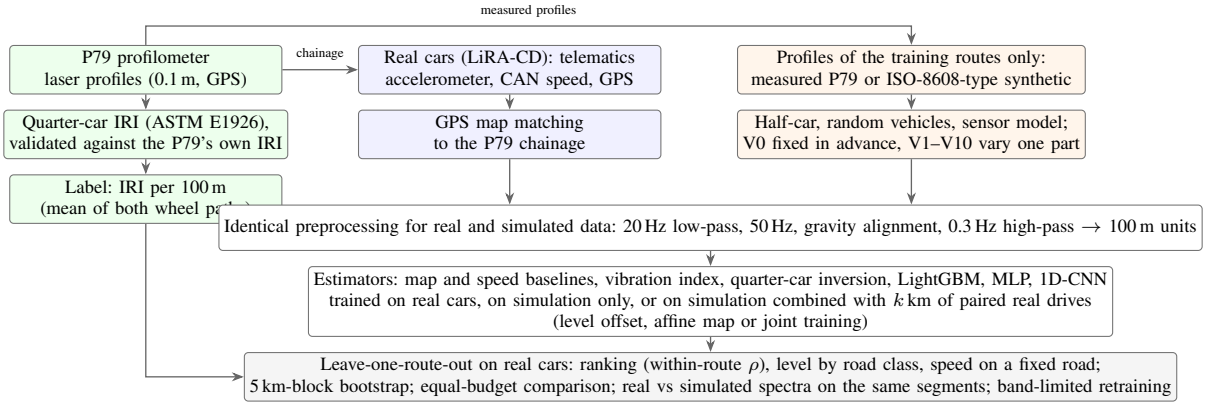


Fig. 1. Study design. Labels are recomputed from the P79 profilometer's laser profiles with an IRI implementation validated against the profilometer's own output; simulated units are labelled with the same code. Real and simulated vibration go through identical preprocessing, and every estimator is evaluated on real routes it never saw; simulated drives never use the profile of the test route.

1.16 m/km at 87 km/h. The same cars drive several routes, so leave-one-route-out evaluation is route-disjoint but not vehicle-disjoint for estimators trained on real data (Section V-A tests the consequence).

Labels. Labels are the IRI per 100 m, mean of both wheel paths, recomputed from the P79 profiles with an implementation of the quarter-car algorithm [16], [2] that reproduces the P79's own IRI ($r = 1.000$ over 1,520 segments; an algorithm check, since both come from the same profiles). As a negative reference we also train on the corpus of our earlier study (19 manually driven trips of 3 BeamNG.tech vehicles on 3 maps), whose labels come from ray-cast game-engine geometry; the curvature-based mesh correction of that pipeline lowers the IRI of the real urban profiles here by 28 %, and its acquisition duplicated 48–68 % of IMU samples (Supplement S2).

Parametric vehicle simulator. A linear four-degree-of-freedom half-car (bounce, pitch, front and rear unsprung masses; validated against a fine-step Runge–Kutta integration, Supplement S3) is driven by a road profile under the front axle and the same profile delayed by the wheelbase under the rear axle, after a 250 mm tyre-envelope moving average; on measured profiles the input is the mean of both wheel paths (no roll), and simulated units are labelled exactly like the real ones. Vehicle parameters are drawn from ranges for passenger cars [36] (sprung mass 1000–1800 kg, wheelbase 2.4–3.0 m, body frequency 1.0–1.6 Hz, damping ratio 0.20–0.45; Table S3), speed independently of roughness (20–110 km/h with fluctuations and braking events). The sensor model applies an optional holder resonance, an anti-aliasing filter, a mounting rotation, irregular sampling, scale and bias errors, white noise, quantisation and sample loss.

Simulator variants. The reference simulator **V0** was fixed before any real test result: 30 random vehicles, measured P79 profiles of the training routes, and a 50/50 mixture of a telematics-like sensor (55–62 Hz, 0.01 g, rigid mount) and a phone-like sensor (90–210 Hz, 0.6 mg) with a holder resonance at 15–40 Hz. To find out which modelling choices matter, we compare it with ten variants, listed in a protocol written after an independent review and before the analyses reported here were run (the zero-shot results of V1–V6 were already known, so this comparison is diagnostic, not confirmatory): the telematics-like half alone (**V1**) and the phone-like half alone (**V2**) of V0; the two missing cells of a 2×2 design, telematics-like with the holder resonance (**V7**) and phone-like without it (**V8**); an ideal sensor (**V3**); a single telematics sensor matched to statistics measured on the unlabelled LiRA-CD streams (**V5**) and its 50/50 mixture with the phone-like sensor (**V6**); one mid-range vehicle instead of 30 (**V4**); and synthetic ISO-8608-type profiles [37] alone (**V9**) or added to the measured ones (**V10**). The simulated drives never use the profile of the test route.

IV. Estimators and Evaluation

Preprocessing and features. Real and simulated accelerometer streams go through the same code: interpolation to 200 Hz, zero-phase 20 Hz low-pass, decimation to 50 Hz, projection on the pass-mean acceleration direction (removing the mounting tilt), and a 0.3 Hz high-pass. LiRA-CD GPS fixes are matched to the P79 track of the same route and direction with a heading constraint. A unit is a 100 m segment traversed once by one pass (≥90 % coverage), described by speed (log mean, coefficient of variation) and vibration features of the dynamic vertical acceleration (log RMS, 95th percentile and maximum of the magnitude, crest factor, kurtosis, band RMS in 0.5–2, 2–5, 5–10 and 10–15 Hz, RMS above 1 Hz, RMS at wavelengths below 5 m).

Estimators. All learned estimators predict log IRI. *Without vibration*: M0, the training median; M0b, a road-class prior that predicts each held-out route with the median IRI of the other route of the same class (needs only a map); M1, LightGBM [38] on the two speed features. *Physics-based*: M2c, a vibration index $\log \mathrm{IRI} = a + b \log(\mathrm{RMS}/v^{\alpha})$ with α fitted; M7, a calibrated quarter-car inversion: the empirical transmissibility of the car and sensor (acceleration spectrum divided by the road displacement spectrum at the pass speed) is estimated on the training routes, a quarter car with sensor

gain is fitted to it, each test unit's road spectrum is obtained by spectral division, and IRI follows from the Golden-Car response to that spectrum (this step reproduces the IRI from true profile spectra with $r = 0.94$ in log scale); M7 is also reported with a log-linear output calibration fitted on the training routes. *Learned*: M3, LightGBM on all features (also without speed inputs); M4, an MLP; M5, a 1D-CNN on the resampled acceleration (the architecture of our earlier study); neural nets are averaged over five seeds. *Simulation-trained*: M3 and M2c fitted only to simulated units of a simulator variant (zero-shot). LightGBM selects one of six fixed configurations by inner leave-one-route-out validation when it is trained on the real passes of whole training routes (M1, M3 and the unit-disjoint check of the supplement); every other LightGBM, including the simulation-trained models, the real-only comparator on k km of paired passes and joint training, uses the first configuration.

Paired data and value of simulation. Paired data are k km of training-route road, drawn as contiguous 1 km blocks (20 random draws per k, identical for every arm so that arms are compared draw by draw), with all real passes over them. LiRA-CD drives each 100 m segment about 9.7 times, so $k = 5$ means 475 segment-passes on average (range 301–640), about 47 km of instrumented driving over 4.9 km of profiled road; to test whether this density is needed, at $k = 5$ each segment also keeps at most $m \in \{1, 2, 5\}$ of its passes, drawn at random. These data are combined with an estimator trained on simulated drives over all training-route profiles in three ways: a level offset (the median log ratio of label to prediction over the paired passes), an affine map in log space, or joint training of LightGBM on the simulated units and the paired passes with equal total weight for the two sets (the "simulation + real" design of our first protocol, reported in our first version for one simulation set and omitted in the second). The real-only comparators are LightGBM and the three-parameter vibration index trained on the same passes. At *equal profile budget*, LightGBM is trained either on the real passes over the k km (which need profiles and paired drives) or on simulated passes over the profiles of the same blocks (which need the profiles only).

Speed on a fixed road. Because each segment is driven about ten times at different speeds, the speed dependence of an estimator can be measured on real roads without confounding by roughness: the within-segment elasticity is the slope of the pass-level log prediction on log speed with segment fixed effects; it should be 0 for a roughness estimator, and $\pm 10\,\%$ between 40 and 100 km/h corresponds to |elasticity| ≤ 0.104.

Simulated vs real vibration. Welch spectra of the preprocessed acceleration (2.56 s windows) are compared between real passes and re-simulated passes of four sensor variants and the ideal sensor over the same P79 segments, on cells of segment $\times$ 10 km/h speed band present in both. For all nine measured-profile variants we compare the band RMS features on the same cells (real/simulated amplitude ratio) and the within-segment speed elasticity of the raw RMS (simulated passes restricted to the speed range of the real passes over each segment). To see which inputs carry the differences between variants, the zero-shot LightGBM of every variant is also retrained without the 10–15 Hz band and, separately, on the five features without content above 10 Hz.

Evaluation and inference. The outer split is leave-one-route-out over {CPH1, CPH6, M3, M13}; both directions of the test route are held out together, and no test-route data enter training, calibration or model selection. The evaluation unit is the 100 m segment, predicted by the mean of its pass predictions. We report MAE (primary), bias and Spearman's ρ (also partial ρ given segment speed); *within-route* values are averages over the four routes, *pooled* values are computed over all 1,678 segments, and every MAE is labelled as one or the other. The *level* of an estimator on a route is the mean over its segments of $\log(\text{prediction}/\text{label})$ (0.1 is an over-prediction of about 10 %), and the level of a road class is the mean of the levels of its two routes. Segment errors are spatially correlated (residual autocorrelation 0.44 at 100 m, 0.11 at 1 km), so paired differences between estimators are bootstrapped over 5 km blocks (40 clusters, 2000 resamples), and a difference is called supported only if that interval excludes zero; segment-, 1 km- and route-level intervals are in Table S7; all intervals are conditional on the trained models, and they use 5 km blocks except the speed elasticities of single estimators (Section V-E) and the spectra (Fig. 3a), which use 1 km blocks.

Evidence classes. The estimators, simulator V0, splits and metrics were fixed before any test result (class C1). Everything else on real data was specified in two later protocols, each written after an independent review and frozen before its analyses were run: the simulator variants, calibration, equal-budget comparison, fixed-road speed analysis, spectra and baselines M0b and M7 in the second; joint training for all variants (restoring the "simulation + real" design of our first protocol), the band-limited retraining, the pass density and further diagnostics in the third. These analyses use routes whose results were already known, and the reviewers had computed some of them, so we label them as re-analyses (R) or diagnostics (E), report every variant and draw, and use none of them to select a configuration; four diagnostics were added to the second protocol after some of its results were known (Supplement S1). The vehicle-number experiment uses freshly simulated test data (S). The protocols were frozen as dated files in our working copy, not registered publicly, so their timing cannot be verified independently.

V. Results

A. Real cars

Table I lists every estimator on the held-out routes (per route: Table S4). A map alone (M0b) or speed alone (M1) gives within-route MAEs of 0.75 and 0.69 m/km. The calibrated quarter-car inversion M7 ranks segments as well as the learned estimators ($\rho = 0.82$) but has a larger error (paired difference to LightGBM $+0.074$ [$+0.039$, $+0.105$] m/km), and its output calibration on the training routes made it worse. LightGBM and the MLP differ negligibly ($+0.004$ [-0.013, $+0.023$]); the 1D-CNN of our earlier study is worse by $+0.045$

TABLE I

REAL CARS (LIRA-CD), LEAVE-ONE-ROUTE-OUT, 100 M SEGMENTS. WITHIN-ROUTE METRICS ARE COMPUTED PER HELD-OUT ROUTE AND AVERAGED OVER THE FOUR ROUTES; "PARTIAL ρ": SPEARMAN CORRELATION WITH IRI GIVEN SEGMENT MEAN SPEED. ΔMAE: PAIRED DIFFERENCE TO THE REAL-TRAINED LIGHTGBM WITH A 95 % CLUSTER-BOOTSTRAP INTERVAL OVER 5 KM BLOCKS (40 CLUSTERS); "ROUTES": NUMBER OF HELD-OUT ROUTES ON WHICH THE ESTIMATOR IS BETTER. NEURAL NETS: 5-SEED ENSEMBLES.

Estimator	Within route			Pooled		ΔMAE vs M3	Routes
	MAE	ρ	partial ρ	MAE	bias	[5 km-block CI]	better
No vibration input							
M0 constant (training median)	1.56	–	–	1.67	−0.64	+1.249 [+1.118, +1.387]	0/4
M0b road-class prior (map only)	0.75	–	–	0.81	−0.20	+0.387 [+0.307, +0.475]	0/4
M1 speed only	0.69	0.39	0.19	0.74	−0.33	+0.324 [+0.252, +0.404]	0/4
Physics-based, calibrated on real training routes							
M2c vibration index	0.62	0.76	0.77	0.67	−0.13	+0.251 [+0.202, +0.306]	0/4
M7 quarter-car inversion	0.46	0.82	0.81	0.49	−0.08	+0.074 [+0.039, +0.105]	0/4
M7 + output calibration	0.54	0.82	0.81	0.58	+0.05	+0.155 [+0.109, +0.201]	0/4
Learned on real cars (3 other routes)							
M3 LightGBM (reference)	0.39	0.83	0.81	0.42	−0.11	ref.	–
M3 LightGBM, no speed input	0.62	0.76	0.72	0.65	−0.24	+0.235 [+0.177, +0.294]	0/4
M4 MLP	0.40	0.83	0.81	0.42	−0.09	+0.004 [−0.013, +0.023]	0/4
M5 1D-CNN	0.43	0.81	0.79	0.47	−0.21	+0.045 [+0.019, +0.075]	1/4
Learned on simulation only (zero-shot)							
V0 LightGBM, measured profiles	0.42	0.83	0.81	0.44	+0.01	+0.019 [−0.003, +0.040]	1/4
V10 LightGBM, measured + synthetic	0.44	0.84	0.82	0.46	−0.04	+0.039 [+0.007, +0.068]	2/4

[+0.019, +0.075], almost entirely on CPH1. LightGBM's MAE is 0.60 and 0.70 m/km on the urban routes and 0.13 and 0.15 m/km on the motorways, or 19 % and 12 % of the mean IRI of each class (Table S6). The same telematics units drive several routes, but LightGBM trained only on units that never drove the held-out route is as accurate as on size-matched random subsets (Table S9; the M3 fold is not interpretable), so vehicle overlap did not measurably favour the real-trained estimators (the within-road split of Bajic et al. [28] is in Table S10).

B. Zero-shot transfer: the ranking transfers, the level does not

Trained only on simulated drives over the profiles of the three training routes, the pre-specified simulator V0 reaches a within-route MAE of 0.42 m/km against 0.39 for the real-trained model. The paired difference, +0.019 m/km, has a 5 km-block interval of [−0.003, +0.040], whereas a segment-level interval, which ignores spatial dependence, would have declared the real-trained model better (Table S7). Both rank segments within routes alike ($\rho = 0.83$ vs 0.83; partial ρ given speed 0.81 vs 0.81). Within-route ranking discriminates little, however, because a Spearman correlation within a route is unaffected by any monotone calibration: the untrained statistic $\mathrm{RMS}/v^{0.5}$ reaches $\rho = 0.82$ (partial 0.81). Paired on the same segments (Table S5), real-trained LightGBM ranks the urban routes better than this statistic (by +0.042 and +0.044) and the motorways slightly worse (−0.020 and −0.023), and V0-trained LightGBM differs from it by −0.015 to +0.038; real-, V0- and V1-trained LightGBM are within 0.05 of it on every route, and so, as point estimates, are LightGBM trained on each of the other variants (largest difference over V0–V8 0.038), the MLP, the CNN and M7 (Table S5). What training has to supply is mainly the level of the mapping from vibration to IRI.

V0's agreement in level is not a robust property of simulating over measured profiles (Table II, Fig. 2). Across the nine variants on measured profiles (V0–V8), the within-route ρ stays between 0.82 and 0.84 and the scale b between 0.62 and 0.71 (real-trained: 0.66), whereas the pooled zero-shot bias ranges from −0.09 to +0.43 m/km. Even V0 matches the level only on average: its pooled bias of +0.01 m/km combines a matched urban level (log level +0.006) with a motorway over-prediction of about 16 % (+0.145; real-trained −0.013 and +0.016). In the 2×2 sensor design, adding the holder resonance changes the bias by −0.35 and −0.36 m/km for the telematics-like and the phone-like sensor, and switching the sensor family by −0.16 and −0.17 m/km without and with holder. The vehicle population moves it as much: with V0's sensor mixture, holder resonance included, but one mid-range vehicle instead of 30 (V4), the bias rises by +0.39 m/km. V0's level thus comes from both its holder resonance, meant to model phone holders rather than the car's telematics unit, and its 30-vehicle population. An ideal sensor (V3) and a telematics sensor matched to the rates, quantisation, noise and tilt of the LiRA-CD streams (V5) both overestimate by about 0.4 m/km. Synthetic profiles alone (V9) still rank segments ($\rho = 0.81$) but overestimate on motorways, and the earlier game-engine corpus does not transfer ($\rho = 0.35$, MAE 1.36 m/km; Supplement S2).

C. Where the simulated vibration differs

On the same segments and speed bands, the real car's vibration differs from all simulated variants in the same way below 10 Hz and differently above (Fig. 3a, Table S21): all variants lack energy at 2–5 Hz (real/simulated power 2.4–3.6) and match it at 5–10 Hz, whereas at 10–15 Hz the real car has 5.2 and 4.9 times the power of the ideal and the rigid telematics-like sensor but only 1.9 times that of the phone-like sensor with holder resonance. Across the nine variants the

TABLE II

SIMULATOR VARIANTS: LIGHTGBM TRAINED ONLY ON SIMULATED DRIVES (ZERO-SHOT), TESTED ON THE HELD-OUT REAL ROUTES. POOLED SEGMENT MAE AND BIAS IN M/KM; LEVEL: MEAN LOG(PREDICTION/LABEL) OVER THE SEGMENTS OF A ROUTE, AVERAGED OVER THE TWO ROUTES OF A ROAD CLASS (0.10 IS ABOUT 10 % OVER-PREDICTION); WITHIN-ROUTE SPEARMAN ρ. RIGHT: POOLED MAE WITH 5 OR 20 KM OF PAIRED DRIVES (ABOUT TEN PASSES PER SEGMENT) USED FOR A LEVEL OFFSET (2.5–97.5 % RANGE OVER THE 20 DRAWS IN BRACKETS) OR FOR JOINT TRAINING; MEANS OVER 20 DRAWS. REAL-ONLY LIGHTGBM ON THE SAME DRIVES: 0.68 M/KM (5 KM), 0.46 M/KM (20 KM). CLASS E EXCEPT V0. SCALE b AND SPEED ELASTICITIES: TABLE S14.

	Zero-shot					+ level offset		+ joint training	
Simulator variant	MAE	bias	level urban	level motorway	ρ	5 km	20 km	5 km	20 km
V0 mixture (reference, pre-specified)	0.44	+0.01	+0.01	+0.14	0.83	0.42 [0.40, 0.45]	0.42	0.45	0.44
V1 telematics-like, rigid	0.68	+0.43	+0.14	+0.40	0.83	0.53 [0.47, 0.60]	0.54	0.47	0.44
V7 telematics-like + holder	0.46	+0.07	+0.02	+0.18	0.83	0.43 [0.40, 0.46]	0.43	0.46	0.44
V8 phone-like, rigid	0.55	+0.27	+0.10	+0.26	0.84	0.45 [0.42, 0.49]	0.46	0.45	0.44
V2 phone-like + holder	0.41	−0.09	−0.03	+0.06	0.83	0.42 [0.40, 0.45]	0.41	0.45	0.43
V3 ideal sensor	0.67	+0.40	+0.13	+0.40	0.83	0.54 [0.48, 0.62]	0.55	0.47	0.45
V5 telematics matched to LiRA-CD	0.69	+0.40	+0.12	+0.42	0.82	0.56 [0.51, 0.64]	0.58	0.47	0.44
V6 matched telematics + phone-like	0.46	−0.04	−0.03	+0.19	0.83	0.45 [0.41, 0.48]	0.45	0.45	0.43
V4 one mid-range vehicle	0.63	+0.40	+0.16	+0.29	0.83	0.44 [0.41, 0.46]	0.43	0.47	0.44
V10 measured + synthetic profiles	0.46	−0.04	−0.04	+0.20	0.84	0.46 [0.41, 0.50]	0.46	0.45	0.43
V9 synthetic profiles only	0.71	+0.36	+0.07	+0.49	0.81	0.75 [0.65, 0.88]	0.77	–	–
trained on real cars (M3)	0.42	−0.11	−0.01	+0.02	0.83	–	–	–	–

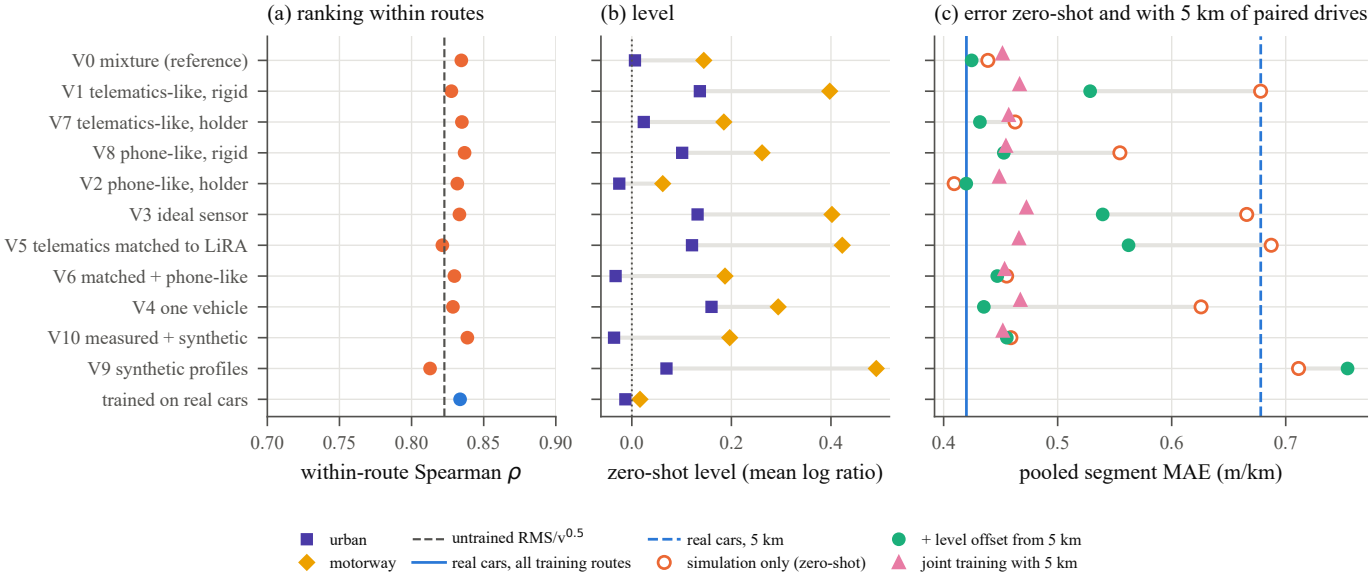


Fig. 2. Simulator variants, LightGBM trained on simulated drives only and tested on the held-out real routes. (a) Within-route Spearman ρ (real-trained model in blue; dashed: the untrained statistic $\mathrm{RMS}/v^{0.5}$). (b) Zero-shot level (mean log ratio of prediction to label, averaged over the two routes of a road class) on urban and motorway segments. (c) Pooled segment MAE zero-shot (open circles) and with 5 km of paired drives used for a level offset (filled circles) or joint training (triangles), means over 20 draws; lines: real-trained on all training routes (solid) or on the 5 km only (dashed).

zero-shot bias orders most closely like the amplitude deficit at 10–15 Hz (Spearman 0.97), but also like those at 5–10 Hz (0.80) and of the overall RMS (0.88), and hardly like those at 0.5–2 and 2–5 Hz (0.22 and 0.25; Fig. 3b, Table S23); a linear fit explains about as much of the bias with the overall RMS ratio as with the 10–15 Hz ratio (R^2 0.78 and 0.71). These associations were found after the fact on variants that are not independent, and V4 departs from them: its 10–15 Hz ratio is close to V0's (1.93 vs 1.86), its bias is not. The deficit is larger on motorway than on urban segments for 9 of 9 variants (Table S22), as is the level error.

To test which inputs carry the differences between the sensor variants, we retrained the zero-shot LightGBM of every variant on the 5 of 13 features that contain no vibration above 10 Hz (speed, speed variation and the 0.5–2, 2–5 and 5–10 Hz bands). Over the eight sensor variants (all but V4), the spread of the mean log level fell from 0.25 to 0.08 (32 %; urban level 49 %, motorway level 34 %) and that of the pooled bias from 0.52 to 0.14 m/km, whereas dropping only the 10–15 Hz band changed every spread by at most 2 % (Fig. 3c, Table S24): the variants differ through all features that carry vibration above 10 Hz, not through the 10–15 Hz band alone. The low-band estimator is not more accurate: for every sensor variant it under-predicts urban and over-predicts motorway segments (log level −0.12 to −0.04 and +0.12 to +0.24), and after a 5 km level offset the pooled MAE of V0–V8 is 0.48–0.57 m/km instead of

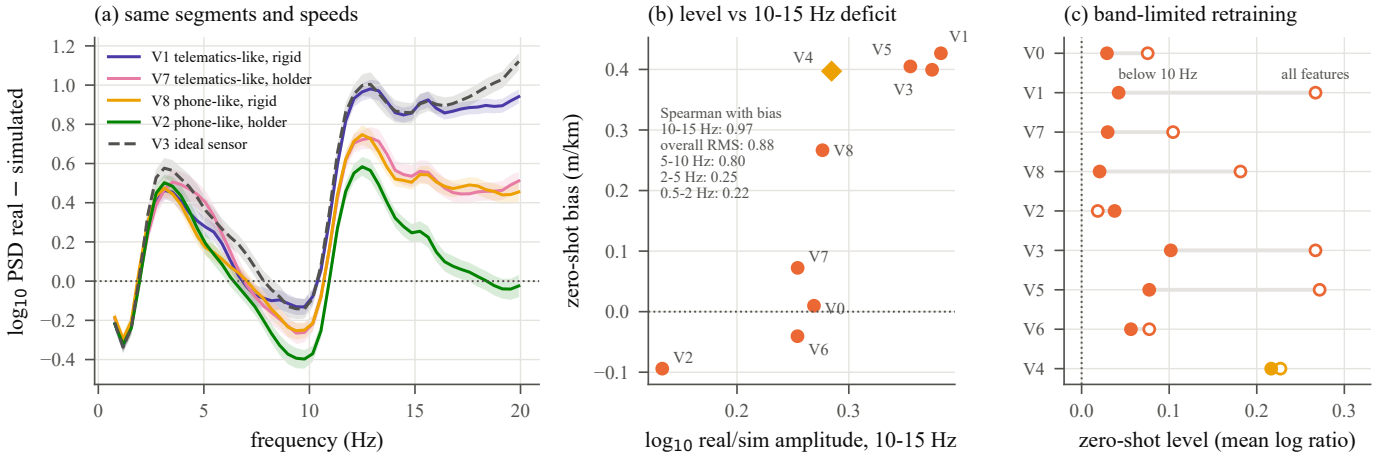


Fig. 3. (a) Mean difference of $\log_{10}$ power spectral density, real minus simulated, on the same segments in the same 10 km/h speed bands (re-simulated passes of four sensor variants and the ideal sensor; bands: 95 % intervals from a 1 km-block bootstrap). (b) Zero-shot bias of the nine measured-profile variants against the real/simulated amplitude ratio of the 10–15 Hz band ($\log_{10}$; diamond: V4, one vehicle), with the Spearman correlation of the bias with the ratio of every band. (c) Mean zero-shot level of each variant with all 13 features (open) and retrained on the five features without content above 10 Hz (filled).

0.42–0.56. V4 stays outside the range of the sensor variants (mean log level +0.22 against +0.02 to +0.10), so the vehicle population sets part of the level independently of the vibration above 10 Hz. The test intervenes on the estimator's inputs, not on the car: it shows where the variant differences enter the estimator, not which physical mechanism produces the real car's extra vibration.

The real car's empirical transmissibility peaks at 12.6 Hz (Fig. S3b), and passes logged at 333 Hz contain more, not less, 10–15 Hz power than passes logged at 52.6 Hz on the same cells (52.6 minus 333 Hz: −0.24 in $\log_{10}$, confounded with the device), so the excess is not an aliasing artefact. Its physical source is not identified: the telematics unit is fixed to the frame in the middle console close to the front axle [15], and the linear half-car omits local modes of the console and its mount, nonlinearity and friction in tyres and suspension, vibration of the electric drivetrain, and roll and body flexibility. A second mismatch is common to all variants: on a fixed road the real car's vibration RMS rises with speed with elasticity 0.60 [0.50, 0.69] on urban and 0.85 [0.74, 0.93] on motorway segments, the simulated vibration over the same segments at the same speeds with 0.92–1.30 and 1.07–1.40 (Table S22); as this mismatch is similar for all variants, it does not explain their different levels.

D. How much paired real data is needed

Fig. 4 and Table III compare simulation with paired real drives on the same 20 draws of k km of training-route road, each with all real passes over it (475 segment-passes on average at $k = 5$); Tables S16 and S17 give every variant. Of the two real-only comparators, the vibration index suits very small samples better (pooled MAE 1.30 vs 1.63 m/km for LightGBM at 1 km). Ranges over draws reflect which k km are drawn; intervals over the test routes are in Table S18.

Equal profile budget (Fig. 4a). Simulated drives over the profiles of the k km are no general substitute for the real drives over them: with 1–2 km, simulated LightGBM beats real-trained LightGBM for every variant but not the real-trained index (better in 3–10 of 20 draws at 1 km), and from 5 km on the answer depends on the variant (at 20 km, −0.02 to +0.24 m/km against real-only LightGBM, largest for V1, V3 and V5, the variants with the largest 10–15 Hz deficit); a level offset from the same k km closes the gap of V4 (+0.01 m/km) but not theirs.

All profiles plus paired drives (Fig. 4b). Trained on simulated drives over all profiled training roads and combined with 5 km of paired drives, every variant beats training on those drives alone (pooled MAE 0.68 m/km), whether the paired drives set a level offset (0.42–0.56 m/km, better in 13–20 of 20 draws) or enter joint training with the simulated units (0.45–0.47 m/km, better in 20 of 20 draws). The test-route intervals exclude zero for every variant and both combinations (differences −0.26 to −0.12 and −0.23 to −0.21 m/km), and within-route MAEs agree (Table III). With more paired data the two combinations diverge. Real-only LightGBM overtakes the level offset of V1, V3 and V5, which is better in only 3–5 of 20 draws at 10 km and in none at 20 km (test-route intervals above zero; differences +0.08 to +0.12 m/km), whereas joint training stays better than real-only LightGBM for every variant up to 20 km (−0.03 to −0.01 m/km at 20 km, better in 16–20 of 20 draws, every test-route interval below zero). Per held-out route it is better in 31 of the 36 combinations of variant and route at 20 km; the other 5 (4 on M3, 1 on CPH6) differ by at most +0.016 m/km. With joint training the variant matters little from 5 km on (the nine variants span 0.02 m/km at 5 and 0.02 m/km at 20 km, against 0.14 and 0.17 m/km with a level offset), but it still matters at 1 km (0.44–0.56 m/km).

Which combination is better depends on the kind of level error. A level offset removes a common level and leaves the difference between the motorway and urban levels, which ranges from 0.09 to 0.30 over V0–V8 (Table II) and orders their MAE after the offset (Spearman 0.88). At 5 km, joint

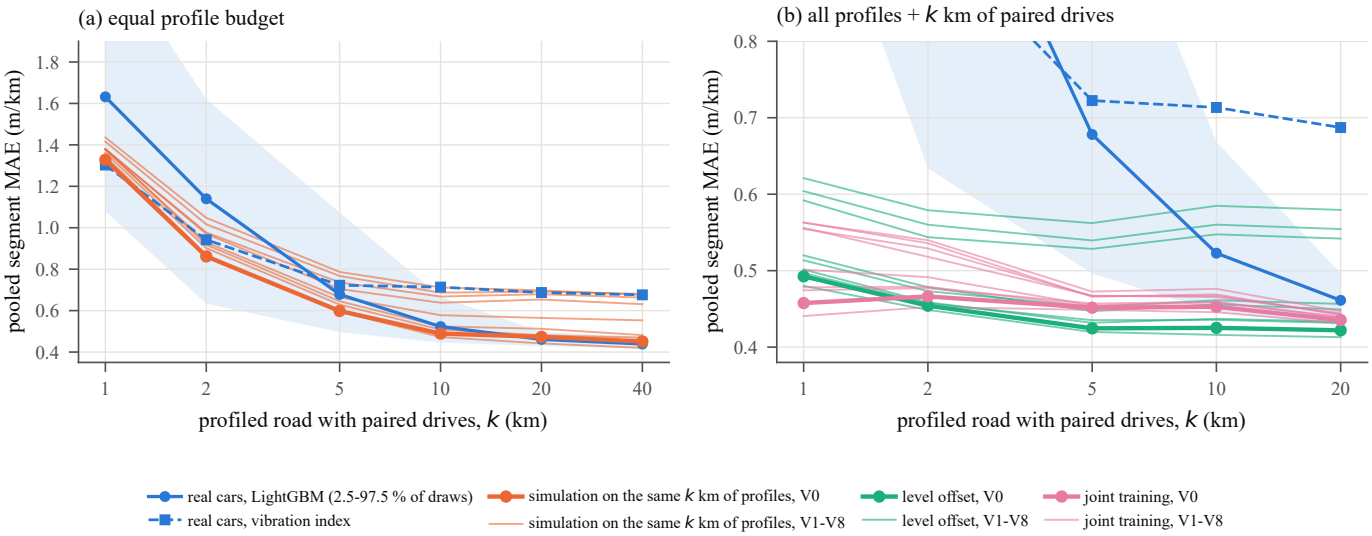


Fig. 4. Simulation against paired real drives on the same 20 draws of k km of training-route road (pooled segment MAE on the held-out routes, means over draws; bold: V0, thin: V1–V8). Blue: real-only LightGBM (band: 2.5–97.5 % of draws) or vibration index (dashed). (a) Equal profile budget: LightGBM on simulated passes over the profiles of the same k km. (b) Simulation over all training-route profiles combined with the paired passes by a level offset (green) or joint training (pink); real-only values at 1 and 2 km lie above the axis (LightGBM 1.63 and 1.14, index 1.30 and 0.94 m/km).

TABLE III

SIMULATION AND PAIRED REAL DRIVES ON THE SAME 20 DRAWS OF k KM OF TRAINING-ROUTE ROAD, EACH WITH ALL REAL PASSES OVER IT (PASSES: MEAN NUMBER OF SEGMENT-PASSES). POOLED MAE OVER ALL SEGMENTS OF THE HELD-OUT ROUTES (BRACKETS: 2.5–97.5 % OF DRAWS) AND WITHIN-ROUTE MAE (MEAN OF THE FOUR ROUTE MAES), IN M/KM. REAL-ONLY: LIGHTGBM OR THE THREE-PARAMETER VIBRATION INDEX ON THE PAIRED PASSES. EQUAL PROFILE BUDGET: LIGHTGBM ON V0 SIMULATED PASSES OVER THE PROFILES OF THE SAME k KM (DIFFERENCE TO REAL-ONLY LIGHTGBM; DRAWS IN WHICH SIMULATION WAS BETTER). ALL PROFILES: SIMULATION OVER ALL TRAINING-ROUTE PROFILES COMBINED WITH THE PAIRED PASSES; RANGES OVER V0–V8 OF THE MEANS, AND OF THE NUMBER OF DRAWS IN WHICH JOINT TRAINING BEAT REAL-ONLY LIGHTGBM.

		Real-only			Equal profile budget (V0)		All profiles + level offset		All profiles + joint training		
k	passes	LightGBM pooled	within	index	pooled	− real (better)	pooled	within	pooled	within	better
1	92	1.63 [1.08, 2.36]	1.572	1.30	1.33	−0.30 (20)	0.48–0.62	0.450–0.576	0.44–0.56	0.415–0.543	20
2	184	1.14 [0.63, 1.62]	1.105	0.94	0.86	−0.28 (20)	0.45–0.58	0.421–0.539	0.45–0.54	0.426–0.519	20
5	475	0.68 [0.50, 1.07]	0.629	0.72	0.60	−0.08 (16)	0.42–0.56	0.395–0.519	0.45–0.47	0.422–0.445	20
10	947	0.52 [0.45, 0.67]	0.487	0.71	0.49	−0.03 (15)	0.42–0.58	0.391–0.537	0.45–0.48	0.418–0.445	18–20
20	1,905	0.46 [0.43, 0.50]	0.433	0.69	0.47	+0.01 (9)	0.41–0.58	0.388–0.532	0.43–0.45	0.405–0.419	16–20
40	3,777	0.44 [0.42, 0.47]	0.412	0.68	0.45	+0.01 (3)	–	–	–	–	–
all		0.42	0.395	–	0.44 (zero-shot)	–	–	–	–	–	–

training minus level offset is −0.096 to −0.062 m/km for V1, V3 and V5, whose class difference is largest (0.26–0.30), and +0.002 to +0.032 m/km for the other six (0.09–0.22). An affine map fitted to the paired passes was less accurate than the better of the two for 9 of 9 variants at 5 km and 9 of 9 at 20 km (Table S17).

LiRA-CD drives each segment about ten times, but with simulation far fewer passes suffice: keeping at most one pass per segment of the 5 km (49 instead of 475 segment-passes on average) hardly changed the combinations with simulation (level offset 0.43–0.55 m/km, joint training 0.44–0.49), which stayed better than real-only LightGBM on the same passes in every draw for every variant, whereas real-only LightGBM degraded from 0.68 to 0.93 m/km (Table S19). The simulation also contributes more than a speed exponent and a slope: a vibration index $\log \mathrm{IRI} = c + b \log(\mathrm{RMS}/v^{\alpha})$ with the car's own speed exponent ($\alpha = 0.59$–0.72, estimated without labels within segments of the training routes) and an affine fit to the same paired passes reaches 0.78 m/km at 5 km and 0.76 m/km at 20 km.

E. Speed dependence on a fixed road

On a fixed road (median within-segment SD of log speed 0.12; Fig. S5, Table S25; intervals of single elasticities from 1 km blocks, of paired differences and robustness rows from 5 km blocks), the real car's vibration RMS rises by +83 % between 40 and 100 km/h (elasticity +0.66 [+0.60, +0.71]). The estimators trained on real cars over-correct for it: LightGBM's estimate falls by −13 % over the same range (−0.15 [−0.19, −0.11]), also without speed inputs (−0.14), the MLP and CNN behave alike (−0.19 and −0.19), and the vibration index over-corrects most (−0.64). Simulation-trained LightGBM V0 changes by −6 % (−0.06 [−0.11, −0.01]); paired on the same passes, its elasticity exceeds that of the real-trained model by +0.09 [+0.06, +0.12] (5 km blocks), entirely on urban roads

(+0.12 [+0.09, +0.15]; motorways +0.02 [−0.06, +0.06]). Across the nine variants the paired difference ranges from +0.00 to +0.21 (interval excluding zero for 8 of 9), and on motorways every estimate falls with speed. These elasticities change little with telematics-unit fixed effects or with the within-pass speed SD as a covariate (Table S26), except that slow urban passes are more dynamic (within-segment correlation of log speed and log speed SD −0.50): with the speed SD as a covariate the real-trained urban elasticity shrinks to −0.06 [−0.16, +0.02], while the paired urban difference stays at +0.12 [+0.09, +0.16]. The speed coefficient of variation, tied to $1/v$ by construction (correlation −0.86 with log speed), absorbs most of the speed variation within urban segments and turns that difference negative (−0.07 [−0.21, +0.01]), so on urban roads speed and speed variability cannot be fully separated. Simulation-trained LightGBM thus over-corrected less than the real-trained model on urban roads in the main design and with the telematics-unit or speed-SD controls, but not with the speed-CV control, so this advantage is not established; simulation did not make estimates speed-invariant, and given the speed-scaling mismatch of Section V-C any difference is a property of the learned estimators, not evidence that the simulator reproduces the car's speed response.

VI. Discussion

What simulated drives over measured profiles provide. Ranking segments within a route turned out to be the easy part: an untrained, speed-normalised vibration RMS comes within 0.05 on every route of real-trained LightGBM, the MLP, the CNN, the quarter-car inversion M7 and simulation-trained LightGBM under every variant, though not of every estimator: the vibration index M2c, LightGBM without speed inputs and the speed-only model rank some route worse than it, by up to 0.10, 0.18 and 0.72 (Table S5). Simulation supplied the mapping from vibration to IRI up to a level that depended on two simulator choices, the sensor path, whose effect reached the estimator through the features carrying vibration above 10 Hz, and the vehicle population; no variant was built to reproduce the absolute gain of this car's vibration path. For a road agency this suggests simulating the car type over roads it has already profiled and combining the simulation with a few kilometres of paired drives, which in our data did not need more than one pass per segment. At 5 km, a level offset was the most accurate combination for the six variants whose level error differed least between road classes, but it fell behind real drives alone from 10 km for the other three; joint training on simulated and paired passes was better than real-only training on average for every variant and every k we tested, though not on every route and draw, and depended little on the variant from 5 km on, which makes it the safer choice when the simulator's fidelity is unknown. Comparing the class levels after calibration, and the real and simulated spectra of the calibration roads, shows which case applies, although the spectral association was found after the fact on nine non-independent variants.

A correction to our earlier explanations. Our first version attributed the transfer of V0 to the breadth of its sensor model, our second to the holder resonance in half of its passes. Neither is sufficient: with the same sensor mixture and one vehicle instead of 30 (V4) the bias is +0.40 m/km, and V0's matched pooled level hides a motorway over-prediction of about 16 %. V0's zero-shot accuracy is a fortunate combination of simulator choices, not a property of randomising over sensors or vehicles, and we do not report it as evidence that a simulator can be used without real data.

Speed. Smooth roads are driven fast, so estimators trained on real cars can use speed as a cue for low roughness, which is consistent with their over-correction on a fixed road; simulation-trained estimators, whose training speeds are independent of roughness, over-corrected less on urban roads in our main design, but on urban roads speed could not be separated from within-pass speed variability (with the speed coefficient of variation as a covariate the difference changed sign, and with the speed SD as a covariate the interval of the real-trained urban elasticity included zero). We recommend reporting such fixed-road elasticities from repeated passes, with the within-pass speed variability as a control, instead of speed sweeps in simulation alone.

Statistics and reference. Segment errors are spatially correlated: 5 km-block intervals of paired differences were 1.9–2.7 times as wide as segment-level ones, and route-level intervals with four routes include zero for most comparisons (Table S7), so the supported differences describe these four routes, not a population of routes, and benchmarks on road data should resample spatial blocks or routes, not segments. All errors are relative to one P79 survey; a second device (ARAN9000) agrees with it in ranking and median level but not on a few urban segments (Table S13).

VII. Limitations

External validity. LiRA-CD contains one car model with one type of telematics unit on four routes in one city region; road class and route are confounded, and with four routes no statement about route-to-route variation is inferential. Whether the simulated mapping transfers up to a level, and how much paired data the level needs, for other cars, sensors, speeds and road networks is unknown. Transfer to other vehicles and to phones in holders was examined only in simulation (class S, Supplement S7): with 1,200 training passes, the error on unseen vehicles exceeded that on the training vehicles by +70 % with 3 and by +16 % with 30 training vehicles. *Evidence class.* Except for the V0 comparison, all real-data results come from analyses specified after the test routes had been used, several after a review had pointed at the result (Section IV); they generate hypotheses rather than confirm them. *Combination methods.* We compared three ways of combining simulated and real data with one learner and fixed settings (equal total weight in joint training, as in our first protocol); fine-tuning of networks, domain adaptation or Bayesian calibration of the simulator might rank differently. *Simulator.* The simulator is a linear half-car without roll or body flexibility, driven by the

mean of the two wheel-path profiles, with parameter ranges for sensitivity rather than a calibrated fleet; its vibration differs from the real car's at 2–5 Hz and above 10 Hz and in its speed scaling. *Other.* M7 is not a reproduction of [12] and the CNN is the architecture of our earlier study without tuning; the reference surveys preceded the car passes by two to fifteen months, segments are 100 m long, and we did not train on real data from several cities.

VIII. Conclusion

Simulated drives over measured road profiles taught gradient boosting to rank the road segments of a real car about as well as real drives did, under every sensor and vehicle model we tried, but an untrained vibration statistic ranks them nearly as well; the informative test is the level. The simulated level was not reliable: it depended on the simulated sensor path, through the features carrying vibration above 10 Hz, and on the simulated vehicle population, and the simulated vibration grew faster with speed than the real car's. With 5 km of profiled road and its paired drives, simulation over all profiled roads beat training on those drives alone for every simulator variant, through a level offset or joint training and even with one pass per segment; with 20 km, joint training kept a small advantage for every variant, whereas the level offset lost it for the variants with the largest vibration deficit at 10–15 Hz. For a new car type this suggests simulating over profiled roads and training jointly with a few kilometres of paired drives, using a level offset only when the simulated and real spectra, and the class levels after calibration, agree. Whether this holds for other vehicles, sensors and road networks requires new paired data, ideally collected after freezing the simulator.

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